

The Centred State and the Surviving Negative Case

Lean-checked rigidity algebra for Erdős Problem #243

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ABSTRACT

Sylvester's sequence $2, 3, 7, 43, 1807, \dots$ satisfies $a_{n+1} = a_n^2 - a_n + 1$ and has reciprocal sum 1. Erdős asked whether a rapidly growing integer sequence with rational reciprocal sum must satisfy that recurrence eventually. We formalise in Lean the integer state system in which the question becomes an assertion about a single centred error.

Writing $D_{n+1} = a_n D_n$, $C_{n+1} = a_n C_n - D_n$ and $E_n = D_n - (a_n - 1)C_n$, we prove the update law $C_{n+1} = C_n - E_n$ and the defect identity

$$(a_{n+1} - (a_n^2 - a_n + 1)) C_{n+1} = a_n^2 E_n - E_{n+1},$$

and deduce that two consecutive vanishing errors with $C_{n+1} \neq 0$ force the exact Sylvester step; in sequence form, eventual vanishing of E together with eventual nonvanishing of C forces the recurrence eventually. Separately we prove the descent that closes the one-sided case: a nonnegative integer state with $C_{n+1} + E_n = C_n$ has $E_n = 0$ eventually. Finally we prove that survival through h forced updates of the normalised constant-negative template depends only on the initial state modulo $(h + 1)!$.

Problem #243 is open and nothing here decides it. The surviving case is exactly the one the descent does not reach, namely a centred state that is negative infinitely often; we neither exclude such an orbit nor construct one. An exact search over initial states below 5000, reduced by the scale equivariance we also prove, produced forced prefixes of length 17 and no longer; that is a computation, not a theorem. The identification of the state system with reciprocal tails is exposition here and is not part of the formalised statements.

1 The problem

Sylvester's sequence $2, 3, 7, 43, 1807, \dots$ is generated by $a_n = a_{n-1}^2 - a_{n-1} + 1$ and has reciprocal sum 1; the recurrence is exactly the statement that the tail beyond a_{n-1} equals $1/(a_{n-1} - 1)$. Erdős Problem #243 asserts that this is the only way a sequence of that growth can have a rational reciprocal sum:

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Problem 1.1 (Erdős #243). Let $1 \leq a_1 < a_2 < \dots$ be a sequence of integers with

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}^2} = 1 \quad \text{and} \quad \sum \frac{1}{a_n} \in \mathbb{Q}.$$

Then $a_n = a_{n-1}^2 - a_{n-1} + 1$ for all sufficiently large n .

See [3, p. 64] and [4, p. 105]; numbering and status follow Bloom’s catalogue [5]. The problem is open. Below we index the same recurrence as $a_{n+1} = a_n^2 - a_n + 1$, which is the shift the formal sources use; the two forms are the same statement.

Write $\text{syl}(a) = a^2 - a + 1$ throughout, the [Sylvester successor](#).

Relation to prior work. Two results bound what is new here, and both are stronger than anything below. Erdős and Straus proved that if $\lim a_n/a_{n-1}^2 = 1$, the reciprocal sum is rational, and a_n does not satisfy the recurrence, then

$$\limsup_{n \rightarrow \infty} \frac{[a_1, \dots, a_n]}{a_{n+1}} \left(\frac{a_n^2}{a_{n+1}} - 1 \right) > 0,$$

where $[a_1, \dots, a_n]$ is the least common multiple [1]. Duverney proved a conditional form of the problem itself: if $\sum_{n \geq 0} (a_{n+1}/a_n^2 - 1)$ converges, then the reciprocal sum is rational if and only if the recurrence holds for all large n [2]. Neither is formalised here, and the results below do not recover either of them. The statement of Problem #243 has also been formalised before, as a conjecture with an unfilled proof, in the *formal-conjectures* collection; that is a statement of the question. No claim of priority is made for anything below.

Why a state system. The hypothesis is asymptotic and the conclusion is exact, so the argument must convert an analytic rate into an integer obstruction. The conversion used here is the classical one: clear denominators along the sequence, so that rationality makes a tail integral, and then centre that integer at the value it would take on Sylvester’s sequence. What remains is a single integer error whose sign controls everything. Section 2 sets this up, Section 3 proves the rigidity step, Section 4 closes the case where the error is nonnegative, and Section 5 reduces the surviving case to a finite-horizon residue computation.

Status of everything below. *Status.* The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no sorry, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Reading map. Sections 2–4 are the rigidity argument and its one-sided closure; Section 5 is the reduction used by the search; Section 6 states what survives. Linked phrases open the corresponding Lean declaration at the pinned source revision fcc93d9ade5e.

Keywords. irrationality; Ahmes series; Sylvester’s sequence; unit fractions; Lean 4. **MSC 2020.** 11J72 (primary); 11B37, 11D68, 68V20 (secondary).

Statement	Status	Treatment here
Problem #243 as stated	Open	Not proved.
Update law $C_{n+1} = C_n - E_n$	Proved here	Proposition 2.1.
Defect identity	Proved here	Theorem 3.1, an identity in $\mathbb{Z}[a, a', D, C]$.
Two vanishing errors force the step	Proved here	Theorem 3.2.
Sequence form	Proved here	Theorem 3.3.
Nonnegative case	Proved here	Theorem 4.1.
Negative case	Open	Section 6; the surviving obstruction.
Scale equivariance	Proved here	Proposition 2.2.
Finite-horizon residue reduction	Proved here	Theorem 5.1.
Longest forced prefix found	Computed	Section 5: length 17, seeds below 5000.
State system equals reciprocal tails	Not formalised	Section 2.
Weighted criterion of Erdős and Straus	Proved elsewhere	Cited, not formalised.
Duverney's conditional characterisation	Proved elsewhere	Cited, not formalised; stronger than anything here.

2 The centred integer state

Fix integers and define, for $a, D, C \in \mathbb{Z}$,

$$\text{den}(a, D) = aD, \quad \text{tail}(a, D, C) = aC - D, \quad \text{ctr}(a, D, C) = D - (a - 1)C.$$

These are the [denominator update](#), the [tail update](#), and the [centred state](#). Given (a_n) we write $D_{n+1} = \text{den}(a_n, D_n)$, $C_{n+1} = \text{tail}(a_n, D_n, C_n)$ and $E_n = \text{ctr}(a_n, D_n, C_n)$.

Proposition 2.1 (update law). $\text{tail}(a, D, C) = C - \text{ctr}(a, D, C)$; that is, $C_{n+1} = C_n - E_n$.

Proof. $C - (D - (a - 1)C) = aC - D$. □

Formalised as the [update law](#). The state therefore moves by exactly the error at each step, which is what makes the sign of E decisive.

The intended reading, which is not formalised. Let $T_n = \sum_{k \geq n} 1/a_k$ and suppose $T_0 \in \mathbb{Q}$. Put D_0 equal to a common denominator and $D_n = D_0 a_0 \cdots a_{n-1}$, and set $C_n = D_n T_n$. Then $C_n \in \mathbb{Z}$ for every n , and from $T_n = 1/a_n + T_{n+1}$ one gets $C_{n+1} = a_n C_n - D_n$, which is the tail update. On Sylvester's sequence $T_n = 1/(a_n - 1)$ exactly, so $D_n = (a_n - 1)C_n$ and $E_n = 0$: the centred state measures deviation from the Sylvester tail identity, and it vanishes identically on the Sylvester orbit.

None of this paragraph is formalised. The Lean module contains no series, no rationality hypothesis, and no growth hypothesis; it proves identities and implications about the integer system above. A reader should treat the identification as motivation for the definitions and not as a checked step.

Proposition 2.2 (scale equivariance). *For every $s, a, D, C \in \mathbb{Z}$,*

$$\text{den}(a, sD) = s \text{den}(a, D), \quad \text{tail}(a, sD, sC) = s \text{tail}(a, D, C), \quad \text{ctr}(a, sD, sC) = s \text{ctr}(a, D, C).$$

Formalised as the [denominator equivariance](#), the [tail equivariance](#), and the [centred equivariance](#). The system therefore depends only on the ray through (D, C) , and a search over initial states may quotient by common scaling. We use this in Section 5; it removed a family of duplicate candidates that differ only by a scalar.

3 The defect identity and the rigidity step

Write $\text{def}(a, a') = a' - \text{syl}(a)$ for the deviation of the next term from the Sylvester successor, the [Sylvester defect](#).

Theorem 3.1 (defect identity). *For all $a, a', D, C \in \mathbb{Z}$,*

$$\text{def}(a, a') \cdot \text{tail}(a, D, C) = a^2 \text{ctr}(a, D, C) - \text{ctr}(a', \text{den}(a, D), \text{tail}(a, D, C)),$$

that is, $\Delta_n C_{n+1} = a_n^2 E_n - E_{n+1}$.

Proof. Both sides expand to $a' a C - a' D - a^3 C + a^2 D + a^2 C - a D - a C + D$. □

Formalised as the [defect identity](#). It is a polynomial identity in four indeterminates, with no hypothesis of any kind.

The identity is the whole rigidity mechanism. The left side vanishes exactly when the sequence takes the Sylvester step or the state C_{n+1} vanishes; the right side vanishes when two consecutive errors do. Reading it in the second direction gives the following.

Theorem 3.2 (two vanishing errors force the step). *Let $a, a', D, C \in \mathbb{Z}$ with $\text{tail}(a, D, C) \neq 0$. If $\text{ctr}(a, D, C) = 0$ and $\text{ctr}(a', \text{den}(a, D), \text{tail}(a, D, C)) = 0$, then $a' = \text{syl}(a)$.*

Proof. By Theorem 3.1 the product $\text{def}(a, a') \cdot \text{tail}(a, D, C)$ is zero, and the second factor is nonzero by hypothesis, so $\text{def}(a, a') = 0$. □

Formalised as the [local rigidity step](#). The hypothesis $C_{n+1} \neq 0$ is not removable: at $C_{n+1} = 0$ the identity gives no information about a' , and indeed a state that has reached zero constrains nothing further.

Theorem 3.3 (sequence form). *Let $a, D, C : \mathbb{N} \rightarrow \mathbb{Z}$ satisfy $D_{n+1} = \text{den}(a_n, D_n)$ and $C_{n+1} = \text{tail}(a_n, D_n, C_n)$ for every n . Suppose that $E_n = \text{ctr}(a_n, D_n, C_n)$ vanishes for all sufficiently large n , and that $C_{n+1} \neq 0$ for all sufficiently large n . Then there is an N with*

$$a_{n+1} = a_n^2 - a_n + 1 \quad \text{for every } n \geq N.$$

Proof. Take N to be the larger of the two thresholds and apply Theorem 3.2 at each $n \geq N$, using the transition equations to identify C_{n+1} with $\text{tail}(a_n, D_n, C_n)$ and E_{n+1} with $\text{ctr}(a_{n+1}, \text{den}(a_n, D_n), \text{tail}(a_n, D_n, C_n))$. □

Formalised as the [eventual Sylvester recurrence](#). Both hypotheses are eventual, and the conclusion is eventual; no compactness or limiting argument is involved, and the two thresholds are merged explicitly.

4 The nonnegative case

Theorem 3.3 needs the error to vanish eventually. When the state and the error are nonnegative that is automatic, by descent.

Theorem 4.1 (descent). *Let $C, E : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $C_{n+1} + E_n = C_n$ for every n . Then $E_n = 0$ for all sufficiently large n .*

Proof. The relation forces $C_{n+1} \leq C_n$, so C is a nonincreasing sequence of natural numbers and is therefore eventually equal to its minimum value; beyond that index $C_{n+1} = C_n$, and the relation gives $E_n = 0$. $\square$

Formalised as the [stabilisation of the nonnegative error](#), on the [eventual constancy of a nonincreasing sequence](#).

The two hypotheses of Theorem 4.1 are exactly the one-sidedness: C and E take values in $\mathbb{N}$. The relation $C_{n+1} + E_n = C_n$ is Proposition 2.1 rewritten, so the theorem is the descent step of every one-sided criterion for this problem. It is stated for abstract $\mathbb{N}$ -valued sequences, and instantiating it at the orbit of Section 2 requires knowing that the state and the error are nonnegative there, which is not proved and is precisely what fails in the open case.

Where the argument stops. If $E_n < 0$ for infinitely many n , then by Proposition 2.1 the state C increases at those indices, and no quantity is known to descend. The reverse one-sided inequality makes the state nondecreasing, which removes the stabilisation argument rather than reversing it. Nothing above excludes that behaviour.

5 A finite-horizon residue reduction

In the surviving case one may normalise the error to a constant negative value, which is legitimate up to common scaling by Proposition 2.2. In that template the next term is forced: at index n its numerator is

$$\text{num}(n, a) = (n + 1)a^2 - (n + 2)a + (n + 3),$$

the [forced numerator](#), and the divisor is $n + 2$. The orbit survives one step when the division is exact, and continues from the quotient. Formally, put $\text{Surv}(0, i, a)$ true and

$$\text{Surv}(h + 1, i, a) \iff (i + 2) \mid \text{num}(i, a) \text{ and } \text{Surv}\left(h, i + 1, \frac{\text{num}(i, a)}{i + 2}\right),$$

the [survival predicate](#). The template itself is a definition in the formal source, not a consequence proved there; the derivation that the surviving case takes this shape is not formalised.

The orbit grows doubly exponentially, so deciding $\text{Surv}(h, 0, a)$ by iteration is expensive. It is unnecessary.

Theorem 5.1 (factorial residue reduction). *For all $h \in \mathbb{N}$ and $a, b \in \mathbb{Z}$, if $a \equiv b \pmod{(h + 1)!}$ then $\text{Surv}(h, 0, a) \iff \text{Surv}(h, 0, b)$.*

Proof. Let $M(0, i) = 1$ and $M(h + 1, i) = (i + 2)M(h, i + 1)$, so that $M(h, i)$ is an ascending factorial and $M(h, 0) = (h + 1)!$. We show by induction on h that $a \equiv b \pmod{M(h, i)}$ implies $\text{Surv}(h, i, a) \iff \text{Surv}(h, i, b)$. The numerator is a polynomial with integer coefficients, so the congruence transfers to $\text{num}(i, a) \equiv \text{num}(i, b) \pmod{(i + 2)M(h, i + 1)}$, and reducing the modulus gives divisibility by $i + 2$ for one exactly when for the other. When both divisions are exact, cancelling the common factor $i + 2$ from the values and from the modulus leaves $\text{num}(i, a)/(i + 2) \equiv \text{num}(i, b)/(i + 2) \pmod{M(h, i + 1)}$, and the inductive hypothesis applies. $\square$

Formalised as the [factorial residue reduction](#), on the [shrinking-modulus transport](#), the [polynomial congruence](#), and the [exact-division cancellation](#); the modulus identifications are the [ascending-factorial form](#) and the [factorial value at the initial index](#).

What the search found. Theorem 5.1 replaces an orbit of doubly exponential integers by a residue computation with a shrinking modulus. Running it over initial states below 5000, quotiented by the common scaling of Proposition 2.2, produced forced prefixes of length 17 and no longer; the leaders include the states 1501, 1723, 2029, 2581, 2971, 3571, 3583, and 4621, each of which fails a later divisibility condition. These are computed values. A prefix of length 17 is not evidence that no infinite orbit exists, and the failure of every tested leader is not a proof that all leaders fail: arbitrary finite deviations can be followed by an exact Sylvester tail, so no finite-prefix rule settles the problem.

6 Complements and further questions

Problem #243 is open. The results above isolate one obligation.

Problem 6.1 (the constant-negative orbit). Show that the divisibility condition sustaining the forced constant-negative template cannot hold at every index, or exhibit an initial state at which it does. By Theorem 5.1 the first h conditions depend only on the state modulo $(h + 1)!$, so the question is whether the shrinking-modulus computation terminates for every residue.

Problem 6.2 (the two published criteria). Formalise the criterion of Erdős and Straus stated in Section 1, and Duverney's conditional characterisation, under explicit analytic hypotheses: the weighted recurrence, the tail error estimate, and the consumer in each case. Both are cited here and neither is formalised; each is stronger than every statement in this note.

Problem 6.3 (the critical boundary). Decide the case in which the normalised under-shoot sits at the reciprocal-index boundary. The constant-negative template of Section 5 lies exactly there, which is why the available rate hypotheses do not exclude it.

The formal content changes the shape of the remaining work rather than its difficulty. The rigidity step is an identity, and it is available at every index; the nonnegative case is closed by descent; and what is left is a single sign condition on an integer sequence, together with the fact that no finite prefix decides it.

Statements and declarations

Artefact and data availability. The [pinned formal-source revision](#) contains the Lean sources, the fixed toolchain, and the library manifest used in the verification. This manuscript provides navigation rather than proof authority.

Declaration of generative AI use. Large-language-model agents were used throughout development to draft and revise prose, formal proofs, and software. The author set the objectives and acceptance criteria, selected and reviewed the claims, and approved the published version. The author assumes responsibility for the accuracy, interpretation, and presentation of the work. Generative systems are production tools; they are not authors and supply no independent authority. Lean checks each proof term against the fixed library version, and the sources linked here contain no proof placeholders and no project-defined axioms; Lean does not authorise the exposition, the citation choices, or the interpretation, for which the author remains responsible.

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A Guide to the formal sources

Each linked phrase opens its Lean declaration at the pinned source revision `fcc93d9ade5e`. The state system, the defect identity, and the descent are in the first module; the residue reduction is in the second. The two are not connected by a formalised derivation: the template of Section 5 is a definition there, and the identification of either module with reciprocal tails is the exposition of Section 2, not a checked statement.

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